To determine suitable ranges of the parameters, I first made a guess using the maximum value in the 3D (log likelihood) grid where the parameters had values $\Omega_m \approx 0.36, \Omega_\Lambda \approx 0.84, H0 \approx 72$. From here I ran the marginalization and viewed the heat maps along with the 1D curve, I observed roughly 4 light areas on each heat map and the peaks of each curve were heavily spiked.

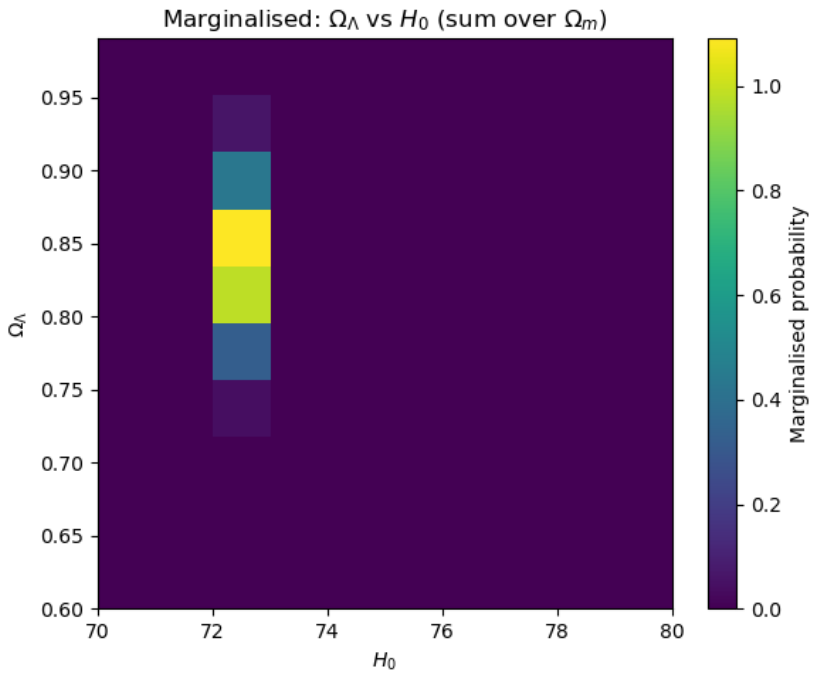


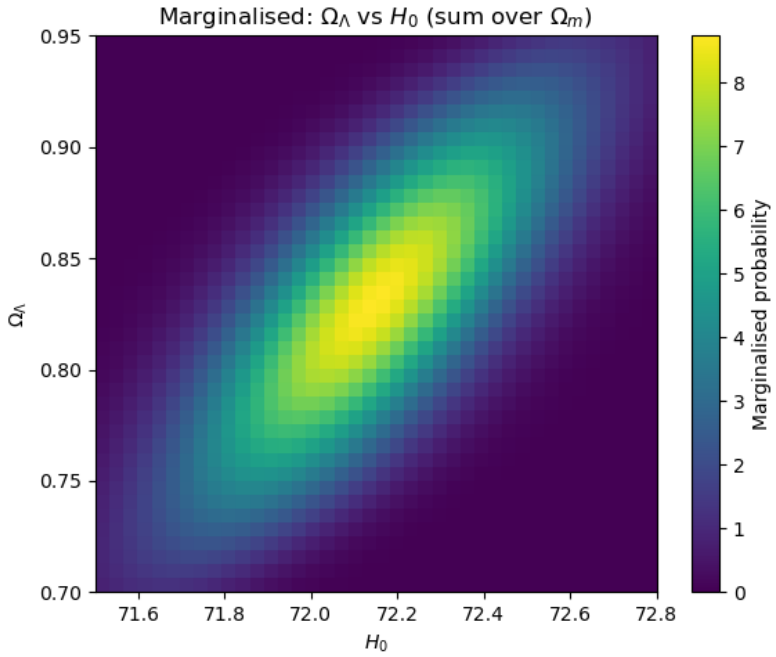
Figure 1: Heat map showing marginalized likelihood of Omega lambda vs H0, displays to wide a range of H0 and Omega lambda such that only a few areas have highly likelihood

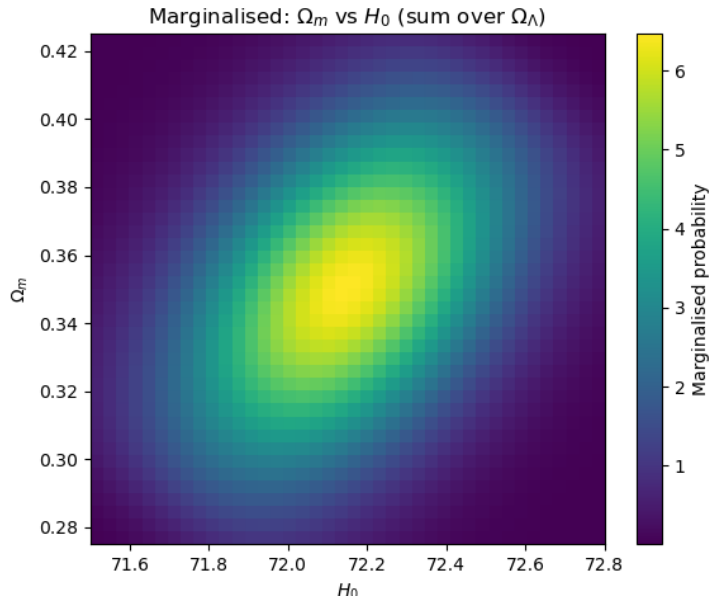
Given that there is a peak in the likelihood (given by the light shaded area), I knew this range included value of each parameter which corresponded to a peak in the likelihood. However, the range was too wide and did not contain fine enough detail (given the 10 steps), so the range must be narrowed down. After trial and error, suitable ranges were determined to include peaks of likelihood (for each parameter) and still show the distributive properties around this peak, shown in the 1D fit by a bell-shaped curve.

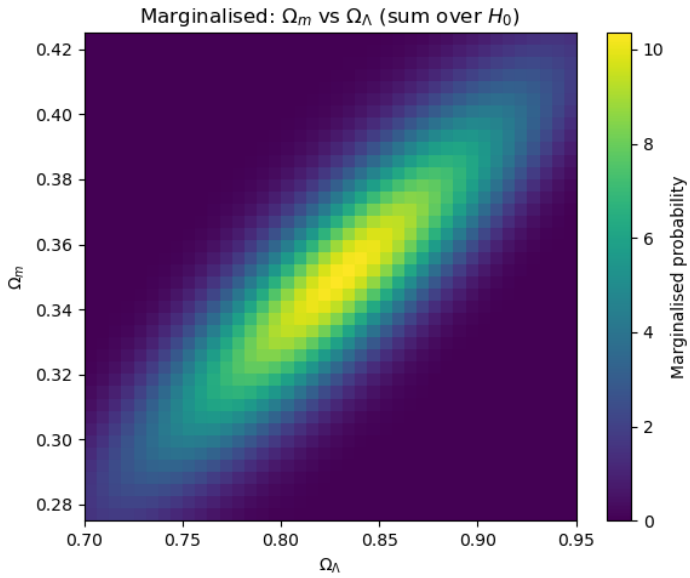
For a finer run and thus more detail in the heat maps the number of evaluations between the minimum and maximum of the range can be increased, this comes with greatly increased processing times since the size of the 3D grid grows very quickly. Steps of 10 were used to determine suitable ranges and processing times was around 5 seconds. Steps will be increased until a suitable tradeoff between processing time and heat map detail is determined (for the purpose of this task a few minutes is an acceptable length of time).

Suitable ranges were determined to be:

$$\Omega_m = [0.275 - 0.425]$$
$$\Omega_\Lambda = [0.7 - 0.95]$$
$$H0 = [71.5 - 72.8]$$

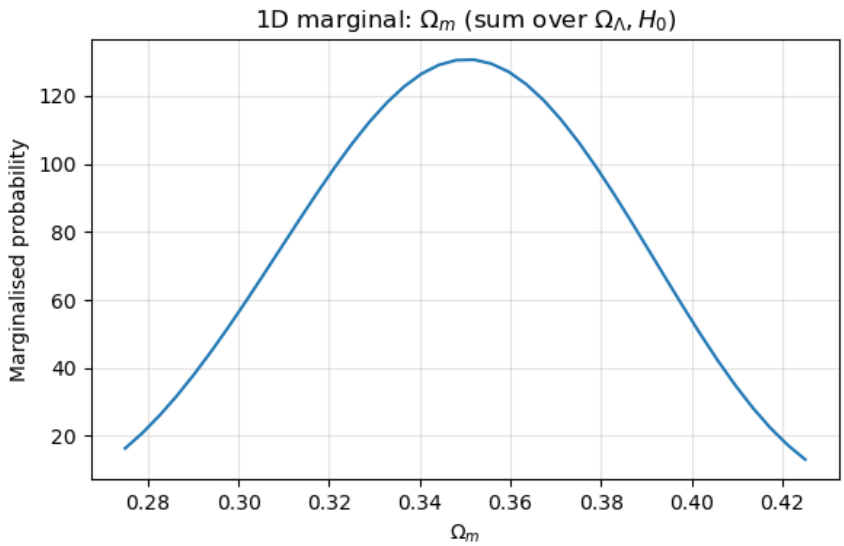




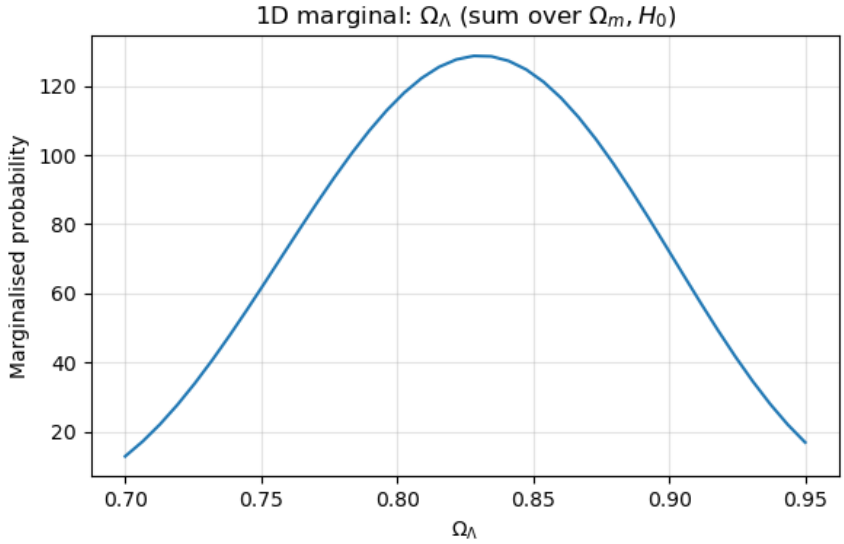


Figures 2-4: Show heat map of likelihoods when one parameter is marginalized, where lighter shaded areas correspond to higher likelihoods within the 2D grid.

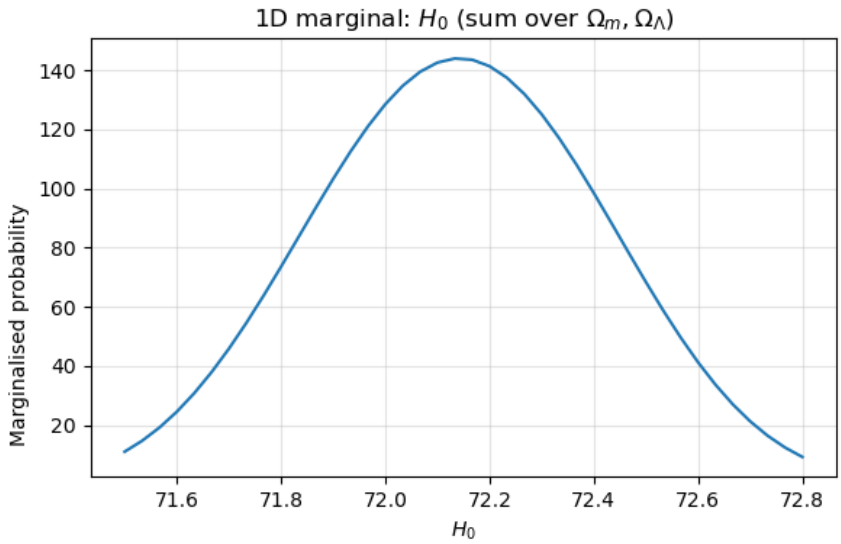
Figures 5-7: Plot of likelihood of only one parameters values while the other two are marginalized. All are smooth peaks within a suitable range that conserve interesting behaviour around peak.



$$\Omega_m = 0.3479 \pm 0.0384,$$



$$\Omega_\Lambda = 0.8246 \pm 0.0657,$$



$$H0 = 72.1231 \pm 0.3037$$

As to comment on the plots above, the heat maps contain clear areas of peak likelihood given by the light colors, and the 1D curves include the peak of marginalized probability for the parameter while leaving out long tails showing the range is closely centered around the point of interest (likelihood peak). Each range had 40 evaluations between

its maximum and minimum values, the total processing time for the heat maps and curve lines to be produced (this includes the grid formation and marginalization) was 2 minutes 45 seconds. Any more evaluation in the range than 40 takes too long (over 3 minutes) and 40 evaluations (per parameter) provides a fine enough heatmap to draw conclusions from for our purposes. The smoothness of the curves can be attributed to the number of evaluations withing the range also, a lower number of evaluation leads to a jagged curve.

Metropolis Algorithm

Using an initial theta guess near the best theta guess from the 3D grid, the convergence of the loglikelihood of theta iterations looks very erratic. This may lead one to believe the Metropolis algorithm is not sampling well, however this can be attributed to the initial theta guess, since every step in any direction around it is likely to be lower and already near the maximum likelihood value (for the best theta), the burn in period is not visible since the algorithm started near maximum theta. This can be deduced from the fact that the loglikelihood ranges from -516 to -524 for all 20000 iterations of theta on the plot.

The initial theta guess was altered, such that every parameter was moderately further away from the best theta guess, so that the burn in period could be visible from the same plot (demonstrating the algorithms ability to sample around the highest likelihood theta iterations).

Also initially, the acceptance rate of proposed theta iteration was $\approx 8.8\%$, this suggests that the sample 'steps' (given in task pdf) are too large and wander out of the accepted likelihood range of iterations with each step (bad proposals). For a higher acceptance rate (better proposals), the size of the 'steps' was reduced to less than half of that of the suggested, $steps\ size\ of\ [\Omega_m = 0.03,\ \Omega_\Lambda = 0.04,\ H0 = 0.12]$. New acceptance rate was 29.2%.

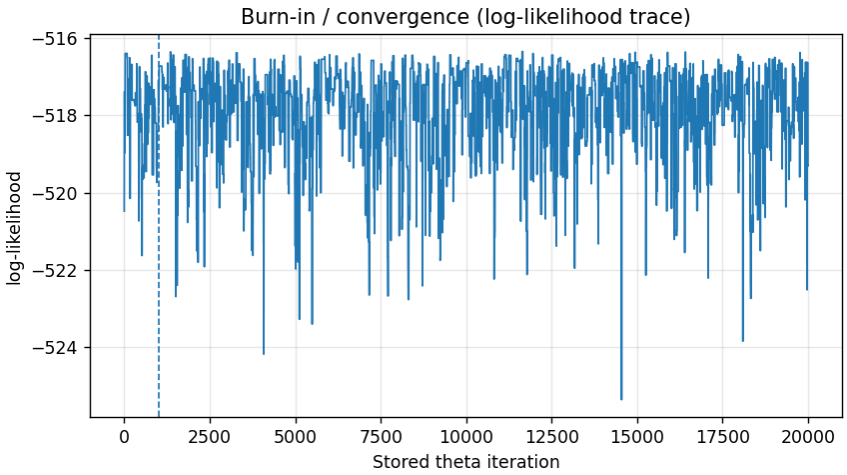


Figure 8: Log likelihood values over evaluated iterations of samples of theta. Plot shows small fluctuations in loglikelihood magnitude around the value determined by the 3D grid (as an initial theta for the Metropolis algorithm).

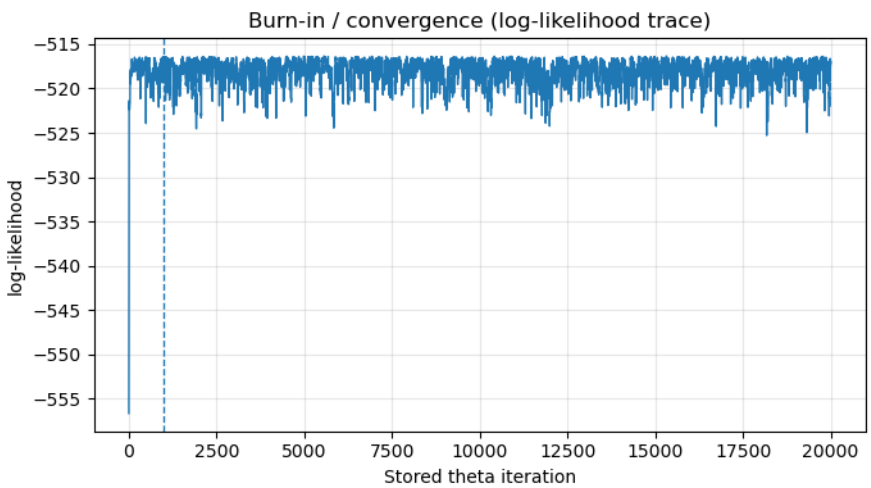


Figure 9: Likelihood plotted against the stored theta iteration. This plot clearly shows the convergence of the loglikelihood (after a burn in period of roughly 200 iterations).

<u>Metropolis Analysis</u>

Using the same initial theta ($[\Omega_m = 0.33,\ \Omega_\Lambda = 0.82,\ H0 = 71]$) and step sizes (as above) in the Metropolis algorithm, the number of evaluated iterations of theta was altered to determine a suitable value (n_steps in the script). The sampler was run for n_steps= 80,000. The loglikelihood trace rises rapidly at the start then fluctuates around a stable value thus indicating convergence. The first 1,000 samples were discarded as burn-in period. The acceptance rate of proposed theta iterations was 29.3%.

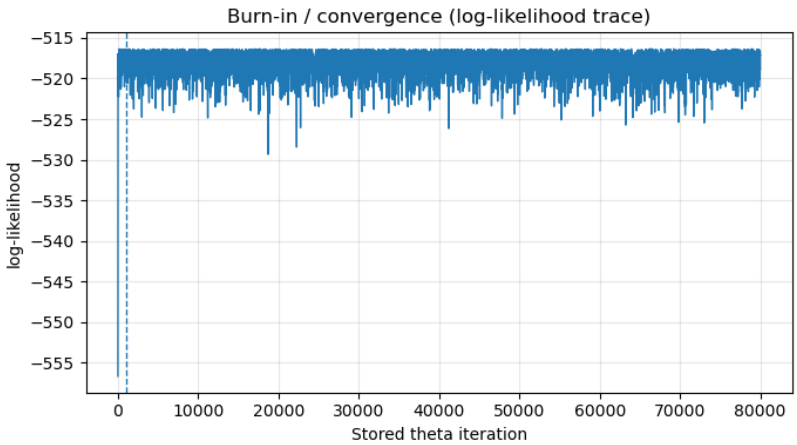
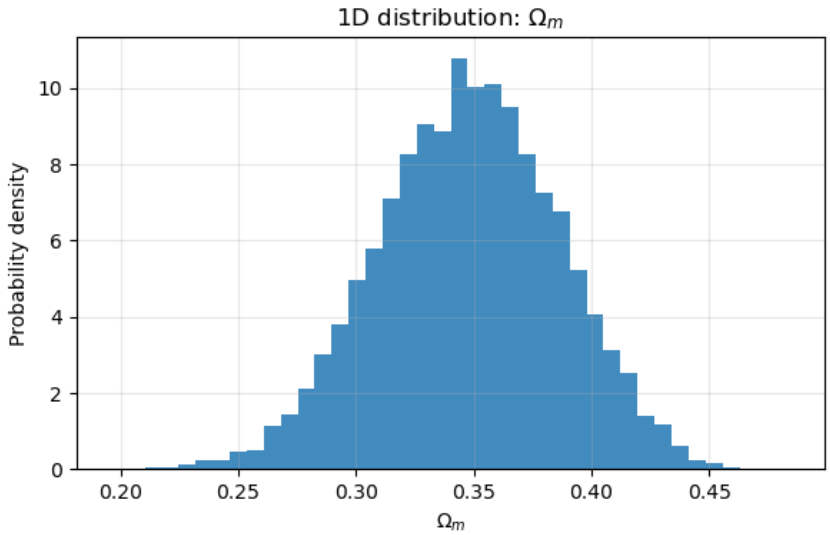
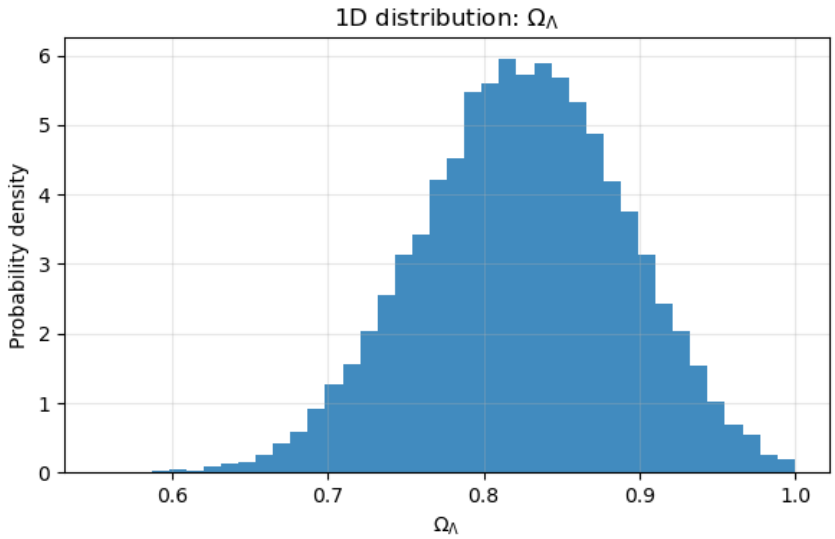


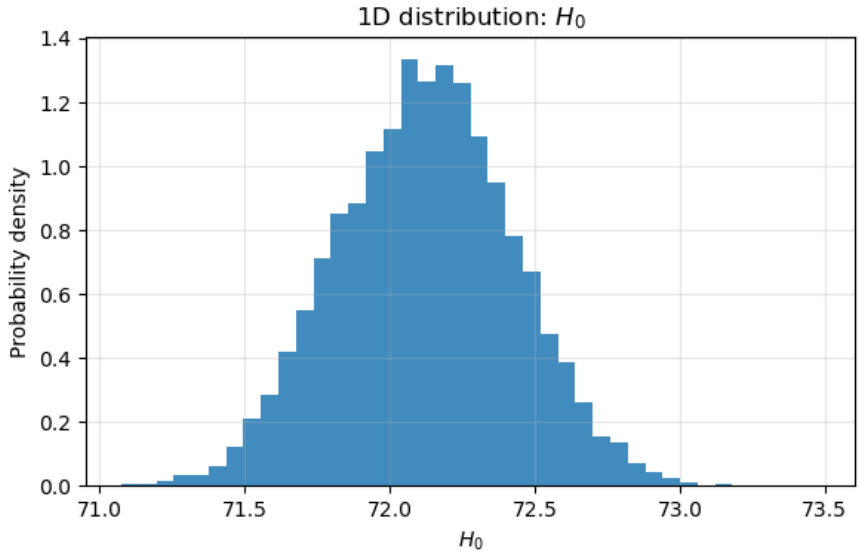
Figure 10: Likelihood plotted against the stored theta iteration.

The 1D histograms distributions are stable once the amount of iterations evaluated surpasses 80,000 (n_steps). Stable meaning after more iterations are evaluated the shape/ smoothness of the gaussian produced is constant in mean and standard deviation.



Figures 10-12: 1D Histograms showing distribution of parameters (Gaussian) obtained from Metropolis analysis after burn in removal.





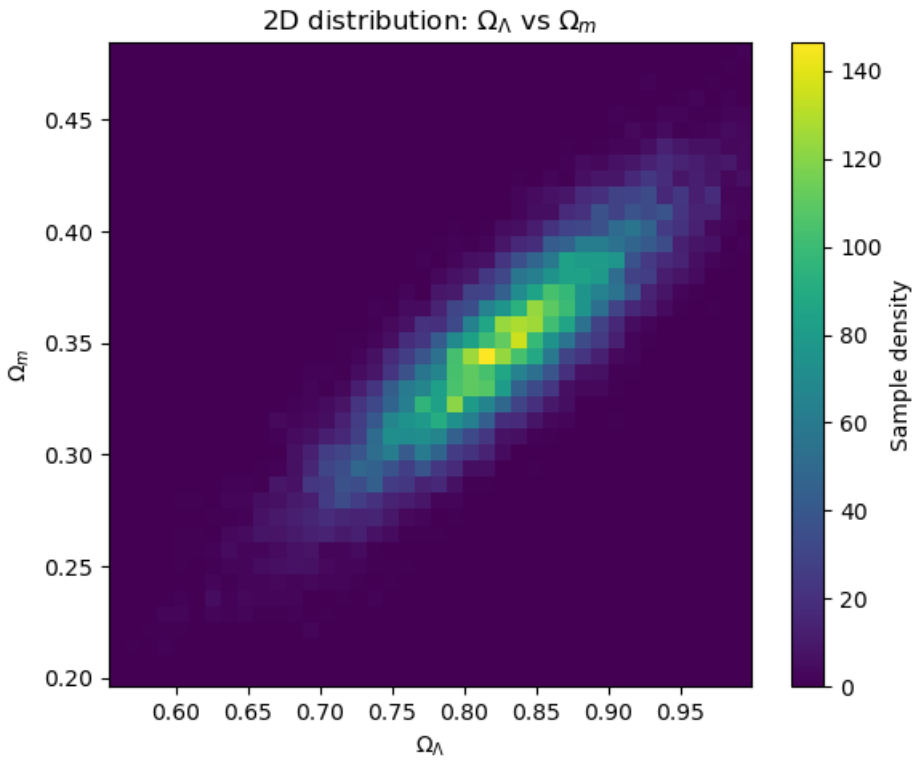
From running the sampler we obtain averages and standard deviations of each parameter as;

$$\Omega_m = 0.3479 \ \pm 0.0384, \qquad \Omega_\Lambda = 0.8246 \ \pm 0.0657, \qquad H0 = 72.1231 \ \pm 0.3037$$

The 2D density plots show clear parameter-parameter correlation:

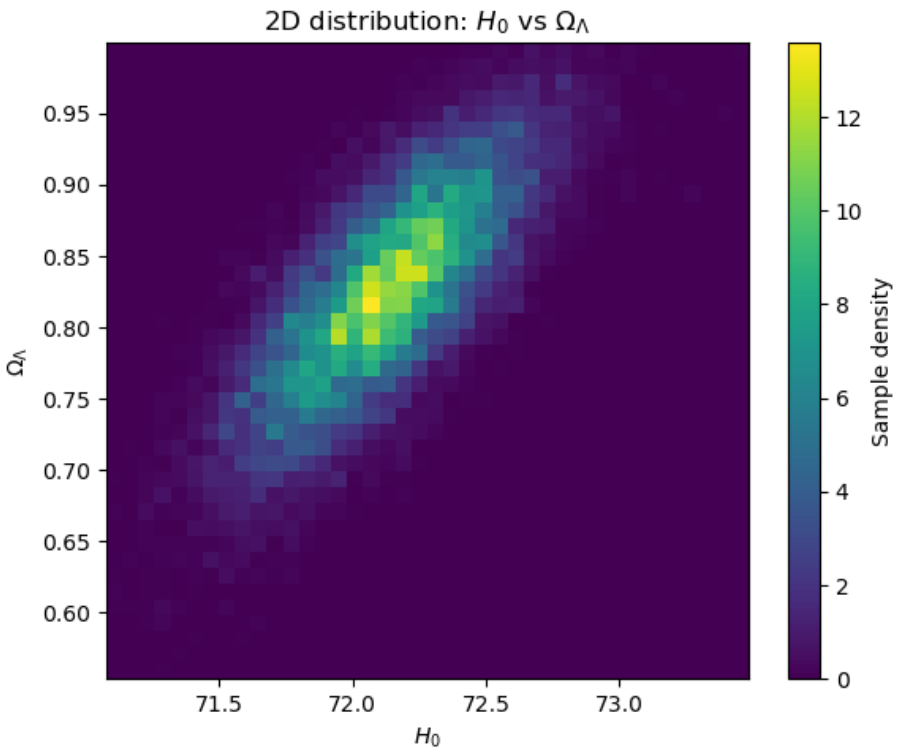
<u>Comparison to Likelihood grid method</u>

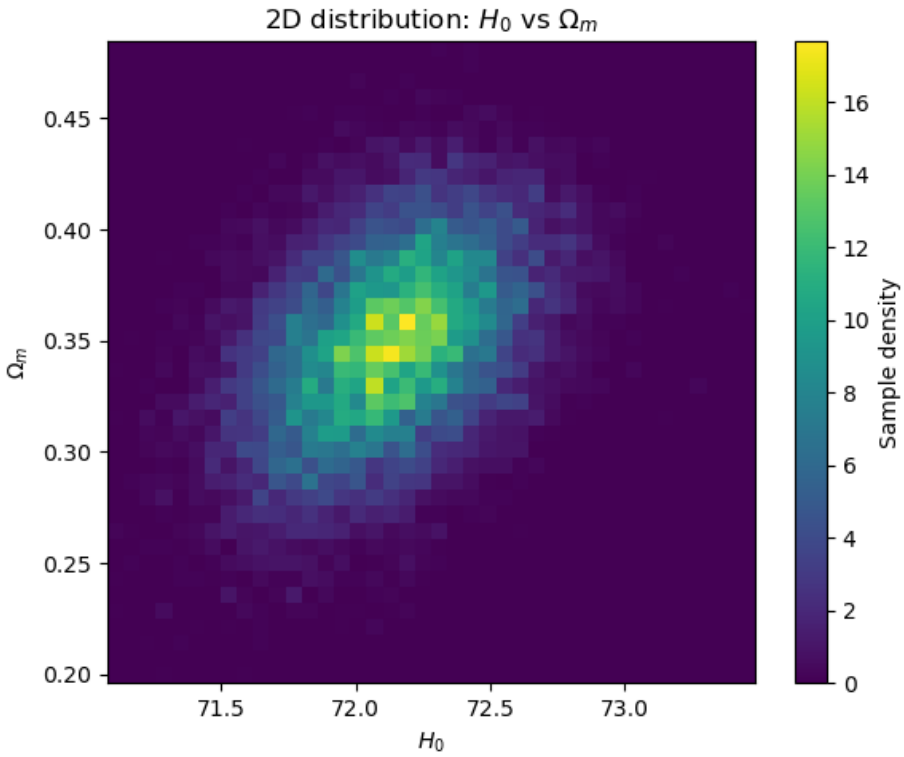
The Metropolis algorithm results are consistent with the likelihood grid method as well as the trends of the marginalized heatmaps. The highest density regions of the histograms (from the Metropolis algorithm) occur in the same part of the parameter space as the heatmaps from the grid method. Although the Metropolis algorithm produced a sample approximation that appears more noisier both approaches identify the same best fit region for theta and are roughly consistent in their uncertainties for $\Omega_m, \Omega_m, and\ H0$.



Figures 13-15: 2D Sample density distribution over 2 parameter spaces at a time. The colour represents sample density and brighter regions indicate higher probability.

While the Metropolis run took 3 minutes 25 seconds, its key advantage is its scalability. The grid method scales exponentially with number of parameters making more than 3 parameters impractical above 3 dimensions. On the other hand, the Metropolis algorithm samples efficiently in higher dimensions, meaning it can handle more parameters without as much processing power.


2D distribution: H_0 vs Ω_Λ
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
Ω_Λ
71.5
72.0
72.5
73.0
H_0
12
10
8
6
4
2
0
Sample density


2D distribution: H_0 vs Ω_m
0.45
0.40
0.35
0.30
0.25
0.20
Ω_m
71.5
72.0
72.5
73.0
H_0
16
14
12
10
8
6
4
2
0
Sample density